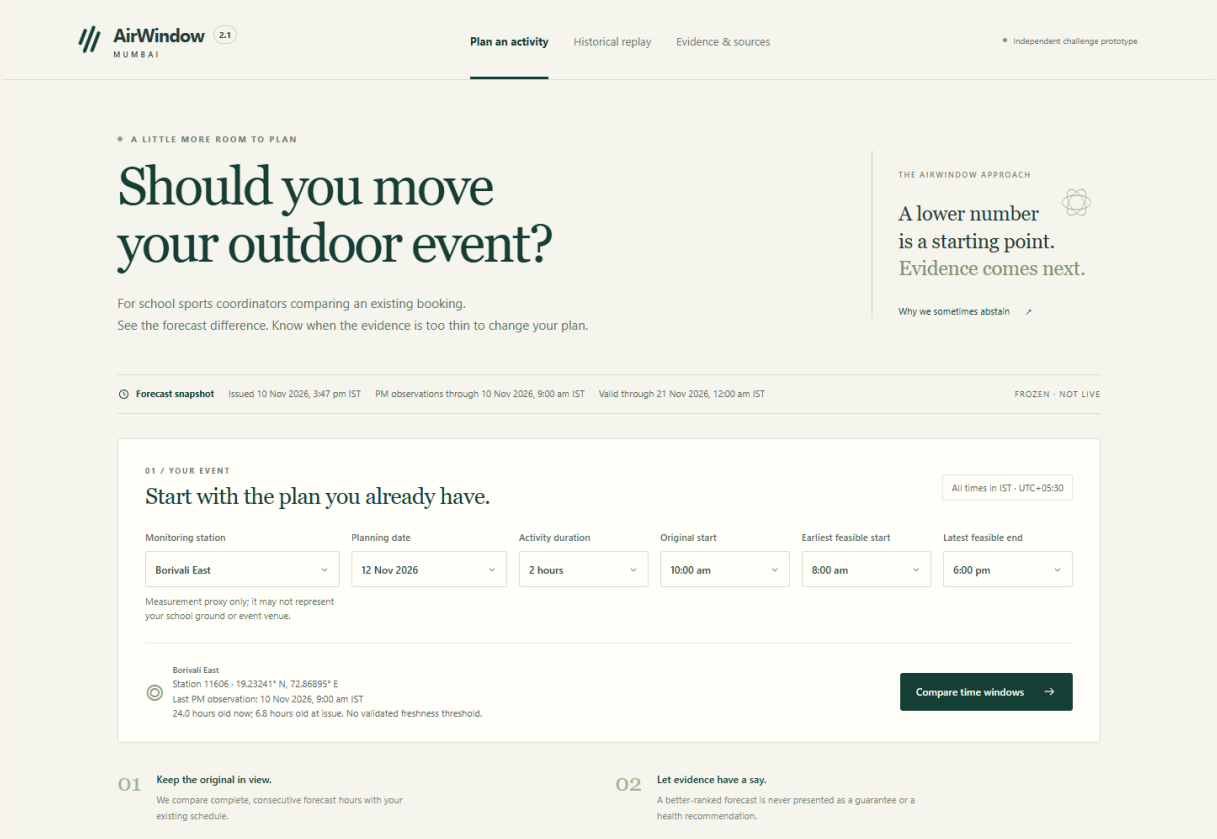
A PRACTICAL DECISION, WITH ITS LIMITS ATTACHED

Should you move your outdoor event?

AirWindow Mumbai

A confidence-aware activity scheduler for school sports coordinators and outdoor-event organisers.



Real application screenshot. Forecast values come from the participant model artifact; no synthetic production data.

Compare your original booking with complete alternatives. See the predicted difference, then see whether historical evidence is strong enough to act on it.

Current evidence is uncalibrated. A lower forecast is not a guarantee, and the product is allowed to abstain.

01 / THE PLANNING WORKFLOW

A booking is the starting point.

A coordinator has a two-hour event already scheduled. It can move between an earliest start and latest finish. AirWindow keeps that original visible while it compares the available options.

Configure	Choose one actual monitor, original start, duration 1/2/3h and feasible range.
Compare	Every candidate requires complete consecutive hourly forecasts. Show mean, maximum, peak hour and difference from the original.
Assess evidence	Show a prominent decision state, comparable support and validation limits. Do not turn numerical rank into certainty.
Replay	Inspect what a genuine historical forecast knew, its selected slot and subsequent actual observations.
Share	Print a one-page event brief with comparison, model version, timestamps and limitations.

Four states, with permission to abstain

ROBUST IMPROVEMENT is reserved for independently validated decision reliability. **TENTATIVE IMPROVEMENT** identifies an unproven numerical alternative. **NO CLEAR WINNER** permits ties or indistinguishable options. **INSUFFICIENT EVIDENCE** makes missing comparable support explicit.

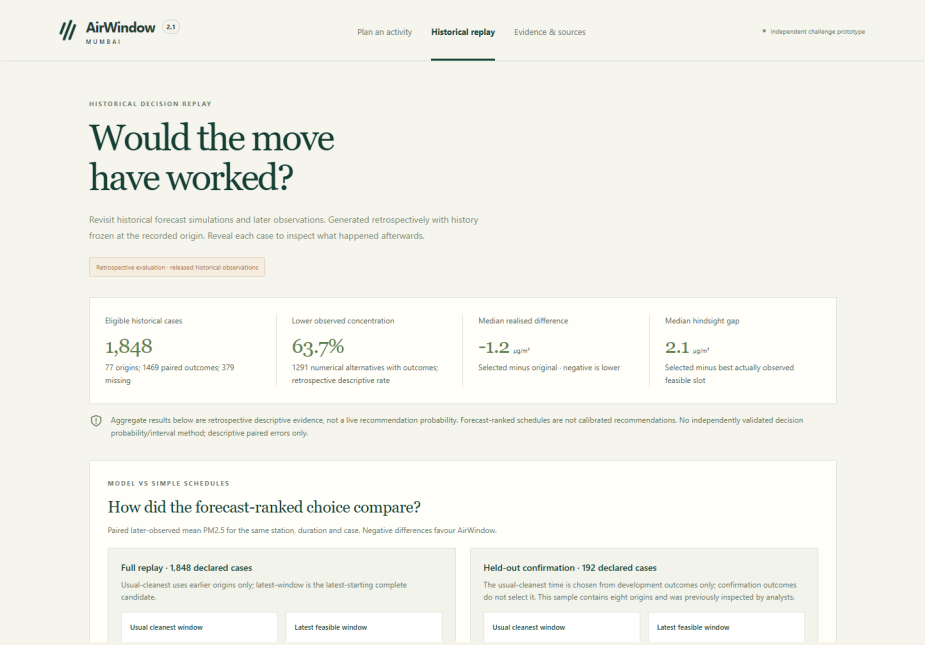
Why this is distinctive

IITM/SAFAR, Google Air View+, Google Air Quality API, London alerts and AirQo already provide forecasts, local information or decision support. AirWindow's narrower contribution is the original-event constraint, paired historical decision replay and honest abstention. This is a product hypothesis, not a first-in-Mumbai claim.

The four supported stations are Borivali East, Malad West, Byculla and Shivaji Nagar. Each is a measurement proxy and may not represent the school ground or event venue. No route, heatmap, medical, safe-window or personal-exposure claim is made.

02 / HISTORICAL DECISION REPLAY

Show the forecast.
Then show what happened.



Real application replay. Aggregate retrospective evidence is visible on entry; selected-case outcomes are revealed explicitly. Subsequent observations remain separate from the live artifact; the table below includes all declared cases.

All declared cases	1,848 cases across 77 forecast origins
Observed coverage	1,469 paired original/selected outcomes; 379 missing outcomes retained
Numerical alternatives	1,291 lower-forecast alternatives with paired observations
Lower afterwards	63.67% of those numerical alternatives had lower later observed PM2.5
Realised difference	Median selected minus original: -1.17 µg/m³ for numerical alternatives
Hindsight gap	Median 2.06 µg/m³ when every feasible observed comparison is complete (1,221 cases); 248 paired cases unavailable

These are descriptive retrospective outcomes, not a probability for today's event, a measured health benefit or a calibrated recommendation. Calibrated recommendations: zero.

03 / REPRODUCIBLE EVIDENCE

The evidence chain is inspectable.

The declared scenario uses a 09:00 original booking, 08:00-18:00 feasible range, durations 1/2/3 hours, both target days, all four stations and every matched OOF origin. Numerical selection occurs before inspecting outcomes. Missing observations remain in coverage.

OOF forecasts were generated retrospectively on 8 November from simulated origins with interval-end PM availability as a proxy. They are not archived real-time vintages.

All-label P_fix RMSE: 757.99 on 11,001 development rows; 15.80 on 1,400 later rows. Eight extremes drive 99.846% of development squared error.

AirWindow versus simple schedule baselines

Paired later-observed mean PM2.5 (AirWindow minus baseline; negative favors AirWindow). Usual-cleanest timing is selected from earlier origins; held-out timing uses development outcomes only. Latest-window uses the latest-starting complete forecast candidate.

Population / baseline	Paired cases / origins	Mean; median delta ug/m^3	95% origin-cluster interval
Full replay / usual cleanest	1,414 / 76	-2.28; 0.00	[-5.72, 0.69]
Full replay / latest window	1,411 / 77	-0.16; 0.00	[-2.34, 2.06]
Held-out / usual cleanest	169 / 8	-0.73; 0.00	[-2.64, 0.54]
Held-out / latest window	170 / 8	-1.01; 0.00	[-2.66, 0.16]

All four cluster-bootstrap intervals include zero. Held-out origins have only n=8 and were previously inspected by analysts. By per-origin mean paired difference (tie tolerance 0.01 ug/m^3), usual-cleanest origins favour AirWindow/baseline/tie: 4/4/0; latest-window: 5/3/0. Equal-origin Student-t sensitivity intervals are [-3.79, 1.23] and [-3.55, 0.72] ug/m^3; both include zero. Exact sign-test p=1.000 and 0.727. Neither method establishes superiority. Cases within origins overlap; paired missing outcomes remain counted in replay coverage.

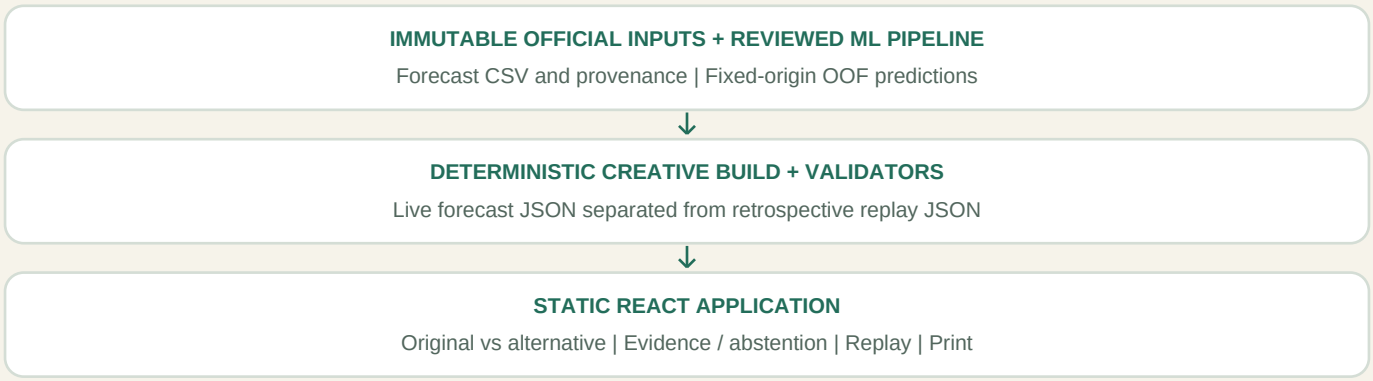
Chronological split	69 development origins; 8 later confirmation origins. Cases overlap within origins and are not independent samples.
Later-origin diagnostic	192 declared cases, 180 paired outcomes, 158 numerical alternatives; 60.76% lower afterwards.
Dependence	Exploratory resampling uses contiguous blocks of four forecast origins, not independent hourly errors.
Calibration limit	Later historical origins had prior analyst exposure. No independent calibration, live probability or uncertainty interval is claimed.
Lead limit	OOF issue equals PM cutoff at 193-240h lead. The current live issue is later than the PM cutoff; matching PM age does not establish matching issue-lead calibration.

Current forecast provenance

Model: P_fix / TASK011 final. Issue: 2026-11-10T15:47:52.944936+05:30. PM-history cutoff: 2026-11-10T09:00:00+05:30. Planning validity horizon ends 2026-11-21T00:00:00+05:30. No weather was used; the prior TASK008 artifact is retained as fallback.

04 / ARCHITECTURE, TOOLS AND HANDOFF

A stable demo, without a live API.



Google tools: Firebase Hosting serves the verified static app at https://airwindow-99085.web.app. The Colab-compatible decision replay notebook was executed locally, not in a verified Google-hosted session; neither is a live application dependency.

Forecast training SHA-256: 8adbcf55315f166d7dc3dc1d3e4d7400651f26c1d8b95ff387eef38f4e70a283
Prediction CSV SHA-256: f450b68212463f64bab67adfb0d98e1c4cce7ee83ec118f2564d38d738e34c13

Deployment status

Static Hosting deployment to the existing airwindow-99085 project succeeded on 11 November 2026 at 21:09 IST. Public URL: https://airwindow-99085.web.app. Signed-out browser verification returned HTTP 200 and checked the forecast, replay (10.5 MB), baseline comparison and preserved fallback JSON against manifest SHA-256 values. Hosting only was deployed; no billing changes were made. The project billing plan was not independently verified. The notebook was executed locally; no cloud execution is claimed.

Working source and reproducibility

Source: github.com/acchasujal/aqi - main branch
Run/build instructions, tests and notebook: apps/airwindow/
Method, red-team review, real screenshots and submission checklist: reports/creative_submission/

Selected primary references

IITM SAFAR project · Google Air Quality API · AirQo AI Platform
London City Hall alerts · Breathe London · Firebase Hosting quotas

Data and use limits

Uses official challenge data and genuine model artifacts; no additional monitoring station was integrated because identity, hourly coverage, provenance and reuse terms were not established. The four supported stations are measurement proxies, not venue-level evidence. No user interviews, measured health benefit, award guarantee or form-processing result is claimed. Consult official CPCB/government guidance for health decisions.

Creative deadline: 13 November 2026, 18:29 GMT / 23:59 IST. Recheck the official form and submit through the human entrant; this package is not submission confirmation.